MGMT 675

AI-Assisted Financial Analysis

Meet your Prof

- At Rice since 2009, in Jones and in Econ Dept.
  - teach core finance, quantitative investments, investments theory, and python for business research
  - to PhD and Masters in Data Science students
- Previously at Northwestern, Indiana, Washington Univ. in St. Louis, and Texas A&M. Associate Dean at Wash U.
- Former and current editor and associate editor of several journals. Two textbooks (derivatives and PhD asset pricing theory). Materials and info at kerryback.com

Motivation for this course

- How will AI affect finance jobs?
- How does this work? (basic template)
  - Python code for financial analysis
  - Create app that encapsulates python code
  - App makes API calls to LLMs to process user's (natural language) input and pass structured version to python code
- Random web cites: 40+% of all new code written by AI, 80+% of developers use AI assistance

Learning objectives

- Learn to use AI to write python code to perform financial analyses
- Get additional practice in financial analyses
- Obtain a basic understanding of
  - Python
  - Making API calls to LLMs
  - Building simple apps

Platforms

- Julius.ai to write and execute python code
- Google Colab as alternative python environment (free)
- VS Code and Cursor as alternative local environments
- Streamlit Cloud for hosting apps (free)
- OpenAI for API calls
- HuggingFace as alternative for API calls

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Deliverables

- 4 individual assignments
- 1 group assignment (last week of class)
- Each assignment is to provide a link to a Julius workflow that will accomplish a specific financial analysis
- Assignments due by midnight on Wednesdays beginning next week

Get a Julius account

- Julius.ai provides a 50% academic discount. Sign up for a free account, then send an email using your Rice email account to team@julius.ai and ask for the academic discount. They will respond with a promo code to use.
- The Lite account ($8 per month after discount) allows 250 messages per month and may be ok.
- Standard account ($18 per month after discount) allows unlimited messages.

Warm-Up

- Ask Julius to plot the payoff diagram of a call option with a strike of 100.
- Ask Julius to use pandas datareader to get the 10-year Treasury yield from FRED, plot it, and save as a jpeg.
- Ask Julius to use yfinance 2.5.4 to get AAPL's closing price from Yahoo Finance, plot it, and save as a jpeg.
- Ask Julius what data it needs to compute the Black-Scholes value of a call option. Supply the data and ask for the value.